

CISC 211 3-1/2 Arithmetic Instructions

Flowchart

Create initial file: Arithmetic.asm

→ Set up initial asm code → section .text → global _start
→ section .data
→ section .bss

Realize I need

to make 4 separate files for each equation.

→ Create new files:

Arith1 - Arith4.asm

Set up each with base initial code from before + fill in rest (.bss & .data)

var	1	2	3	4
4				
5				
6				
7				

initialize vars to 4 for 32-bit ints

→ Arith1.asm

add multiplication ops

↓
Arith2.asm

add sum/add ops

↓
Arith3.asm

add add & mult ops (in proper order)

Arith4.asm

- re-read lecture notes on division closely
- calc denominator and store in ebx (temp)
- calc numerator and store in eax
- How to divide?

- IDIV for signed ints

- look up how to 32 bit divide using IDIV

↳ Stack Overflow.com

↳ Use CDQ on line prior

↳ Then idiv ebx

quotient remainder (should be zero if result is an int)
EDX:EAX

→ Compile time

./run.sh Arith1 → nasm: command not found

re-install nasm → done, try again, nothing so I continue to Gdb

→ gdb: command not found, re-install gdb, run again, no de-bugging symbols? ignore, type gdb commands & watch... They all work!